

The First Marginal of a Transport Plan

Let

$$\gamma \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$$

be a transport plan. The projection onto the first coordinate is the map

$$\pi_x : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}^d, \quad \pi_x(x, y) = x.$$

The condition

$$(\pi_x)_\# \gamma = \mu$$

means that if we ignore the destination coordinate y and keep only the source coordinate x , then the resulting distribution is μ .

Explicitly, for every Borel set $A \subseteq \mathbb{R}^d$,

$$\mu(A) = (\pi_x)_\# \gamma(A) = \gamma(\pi_x^{-1}(A)) = \gamma(A \times \mathbb{R}^d).$$

Thus the total amount of mass in the plan whose source lies in A must equal the original mass $\mu(A)$.

Finite Example

Suppose

$$\mu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$$

and define a transport plan

$$\gamma = \frac{1}{2}\delta_{(0,2)} + \frac{1}{2}\delta_{(1,3)}.$$

Then

$$(\pi_x)_\# \gamma = \frac{1}{2}\delta_{\pi_x(0,2)} + \frac{1}{2}\delta_{\pi_x(1,3)} = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1 = \mu.$$

The first marginal condition says that the plan starts with exactly the source distribution μ .